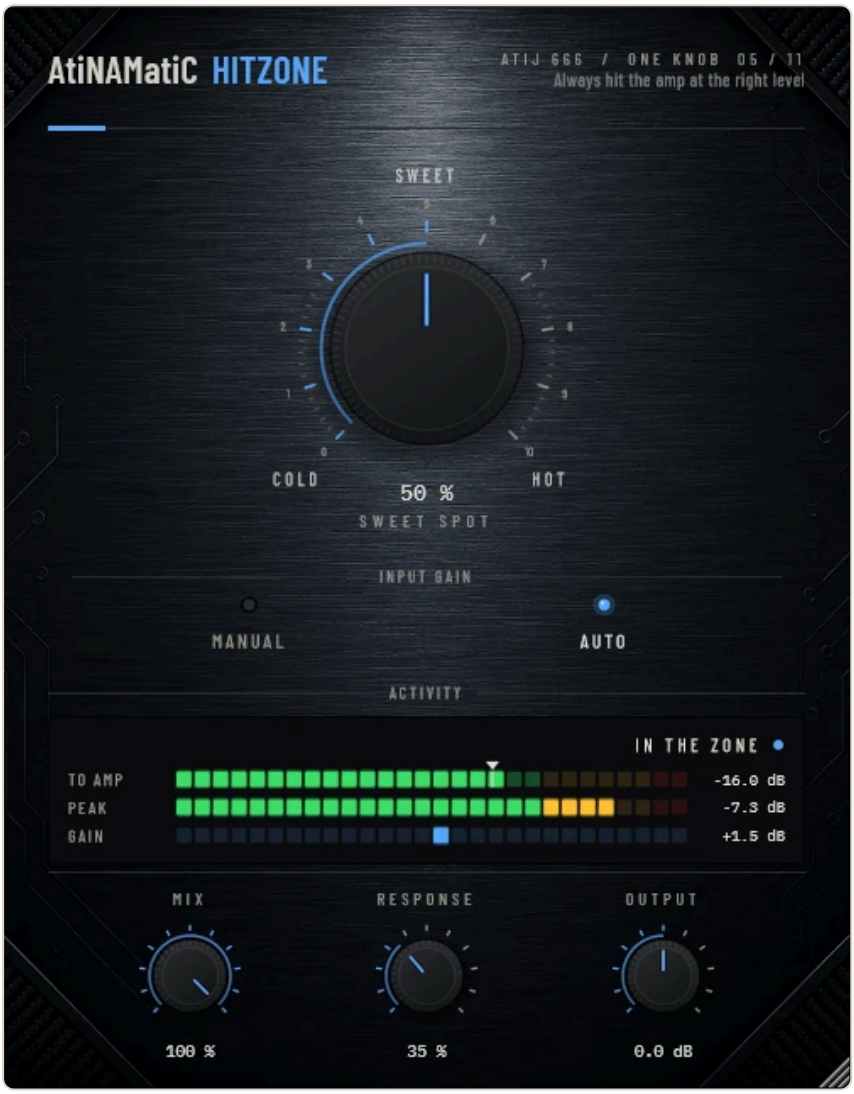
AtiNAMatiC-HitZone — User Manual

Version 1.0.0 · ATIJ STUDIO · Windows · macOS · Linux · VST3 · CLAP · LV2 · AU (macOS) · Free and open source (GPL v3)



What's new in v1.0.0

- First public release, now **free and open source** (GPL v3).
- New look, shared across the whole ONE KNOB Series.
- **Every plug-in format:** VST3, CLAP and LV2 on Windows, macOS and Linux, plus AU on macOS. (The standalone app has been retired: these work best inside your DAW.)
- New **ONE KNOB Series installers** for Windows and macOS: tick the plug-ins and formats you want, skip the rest. Linux gets an install script that asks the same questions.

What's new in v0.2.0

- **Auto gain fixed.** The auto loop compared the *input* level with the target and kept adding the same error every block, so it ran to the ±24 dB limit instead of settling. It now regulates the level *after* its gain, settles on the -16 dBFS RMS target, and holds its gain through silence instead of winding up between phrases.
- **Peak protection.** A loud hit after a quiet passage is caught instantly (block-peak ceiling + near-instant gain drop), with a smooth safety knee above -3 dBFS in Auto mode.
- The status label and meter now show the level *going into your amp* (after HitZone's gain), which is what "in the zone" means. The SWEET SPOT tone shelves now reach their setting immediately (they used to take tens of seconds).
- **New interface** — machined knob with a printed 0–10 scale, hardware-style mode switches that follow automation and preset recall, consistent value text ("72 %", "+1.5 dB"), resizable window (75–150 %, remembered per session). Double-click any knob to reset it; hold **Shift** while dragging for fine adjustment.

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1. What it is

AtiNAMatiC-HitZone is an intelligent input-level conditioner designed to sit **before** Neural Amp Modeler (NAM) plugins in your signal chain. Its job is simple and critical: make sure the signal hitting your NAM model is always at the exact level those models were trained at.

NAM models are typically captured with peaks around –18 dBFS and an RMS around **–16 dBFS**. Feed them too quietly and the amp sim sounds dull, thin, and lifeless. Feed them too loud and they compress, clip, or distort in unpleasant, unintended ways. HitZone watches your input level in real time, adjusts gain automatically, and tells you in plain language: **Too Cold**, **Perfect**, or **Too Hot**.

Beyond gain management, the **Sweet Spot** macro tunes the tonal character of the conditioned signal — from bright and tight (COLD side) to warm and driven (HOT side) — using gentle EQ shelves and a touch of asymmetric tube-style saturation.

In short: AtiNAMatiC-HitZone takes the guesswork out of driving NAM correctly.

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2. Signal chain placement

```
Guitar → ★ AtiNAMatiC-HitZone ★ → NAM Plugin → Cab / IR → Post-FX → Master
```

Always place HitZone directly before your NAM model, never after it. The level meter and auto-gain are calibrated to NAM's training level. Placing it after NAM has no benefit and may introduce unwanted colouration.

If you use a noise gate, place it **before** HitZone so the gate's closed state does not confuse the auto-gain with near-silence readings.

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3. The Controls

SWEET SPOT (large central knob)

The master macro knob. Turning it from COLD to HOT simultaneously adjusts:

- **Tonal EQ** — COLD side lifts 4.5 kHz (+2 dB high shelf) and slightly cuts 180 Hz (–2 dB low shelf) for a bright, present character. HOT side does the opposite: +3 dB warmth at 180 Hz and –2.5 dB at 4.5 kHz for a thick, vintage feel.
- **Saturation** — below centre the signal is clean; above centre a gentle asymmetric tube-style soft-clip (2nd-harmonic bias) blends in progressively, reaching full drive at the HOT extreme.

Most players find their sweet spot somewhere between 35 % and 65 %. Highly distorted amp models benefit from the COLD side (keeps things tight); cleaner and crunch models often respond well to the HOT side.

Default: 50 % (centre — flat EQ, no saturation)
Double-click to reset to default.

AUTO (toggle, centre)

Enables real-time automatic gain adjustment.

State	Behaviour
ON (default)	HitZone continuously monitors RMS level and applies up to ±24 dB of gain to keep your signal in the sweet zone (–20 to –12 dBFS, ideal –16 dBFS).
OFF	Auto-gain is disabled. The **GAIN** knob (replaces RESPONSE in the bottom row) lets you dial in a fixed input gain manually.

Auto mode uses an **asymmetric tracking algorithm**: it pulls gain down quickly when the signal is too hot (to protect the NAM model from clipping), but brings gain up slowly when too cold (to avoid level-pumping on transients and note attacks).

RESPONSE (bottom row, centre — when AUTO is ON)

Controls how fast the automatic gain tracks changes in input level.

Position	Up speed	Down speed
0 % (Slow)	~1.5 dB/sec	~6 dB/sec
50 % (default)	~6 dB/sec	~16 dB/sec
100 % (Fast)	~18 dB/sec	~60 dB/sec

Slow response is natural and transparent — good for playing through full songs. Fast response is useful when switching between quiet clean passages and high-gain riffs in the same session.

A **safety stomp** is always active regardless of Response: if the peak level approaches –3 dBFS (dangerously close to clipping the NAM model), the gain is pulled down at the fastest available rate immediately.

Default: 35 %
Double-click to reset to default.

GAIN (bottom row, centre — when AUTO is OFF)

Fixed input gain trim. Range: –24 dB to +24 dB, default 0 dB.

Use this when you want total manual control — for example when recording automation, when your DAW chain already includes level management, or when tracking the gain for a specific guitar/pickup combination.

The **Gain Readout** (small number to the right of the status label) shows the currently applied gain in dB — in Auto mode this reflects the live auto-gain value; in Manual mode it reflects the GAIN knob.

Double-click to reset to 0 dB.

MIX (bottom row, left)

Wet/Dry blend. 0 % = fully dry (bypassed), 100 % = fully processed.

At values below 100 % the processed and original signals are summed. This is useful for parallel-processing workflows: run HitZone at 100 % for the NAM chain and blend in a small amount of unprocessed signal for transient clarity.

Default: 100 %
Double-click to reset to 100 %.

OUTPUT (bottom row, right)

Final output level trim, applied after all processing. Range: –24 dB to +24 dB, default 0 dB.

Use this to compensate for any net level change introduced by the EQ shelves or saturation, or to match levels between HitZone on/off.

Double-click to reset to 0 dB.

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4. Status indicators

Status label (centre of UI)

Displays the current input-level state in real time, updated at 20 Hz:

Label	Meaning	What to do
Too Cold (blue)	RMS below –21 dBFS — signal is too quiet for NAM	Increase gain at source, or turn up GAIN knob (manual mode)
Perfect (green)	RMS between –21 and –11 dBFS — NAM sweet zone	Nothing — you're in the zone
Too Hot (orange/red)	RMS above –11 dBFS — signal risks clipping the NAM model	Reduce gain at source, or use AUTO

Status uses hysteresis to prevent rapid flickering at zone boundaries: the indicator does not switch back to Perfect until the signal has moved 0.5–1 dB clear of the threshold.

In **AUTO** mode, HitZone drives itself to maintain "Perfect" status. The label may briefly show Too Cold or Too Hot during fast transient changes.

Gain Readout (small number, right of status)

Shows the currently applied input gain in dB (up to one decimal place). In AUTO mode this is the live auto-gain value; in MANUAL mode it mirrors the GAIN knob. Useful for setting a fixed trim: switch AUTO off, watch the readout until it stabilises in manual mode, then use that value as your GAIN knob reference.

Level Meter

A horizontal RMS level meter spanning the COLD–HOT range, positioned between the Sweet Spot knob and the status row.

Element	Meaning
Shaded band	Sweet zone (–20 to –12 dBFS) — target region
Filled bar	Live RMS level
Peak indicator	Short-term peak envelope
Peak-hold dot	Holds the highest peak seen, decays slowly (1.5 s)

The COLD and HOT labels at the left and right of the knob area indicate the tonal character of the Sweet Spot position, **not** the current level status. Level status is indicated by the status label.

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5. System requirements

	Minimum
OS	Windows 10 (64-bit) or later
CPU	Any modern x86-64 processor
RAM	4 MB per instance
DAW	Any VST3-compatible host (64-bit), or CLAP-compatible host
Sample rates	44.1, 48, 88.2, 96, 176.4, 192 kHz
Buffer sizes	Any (zero-latency processing)

6. Installation

Windows (64-bit)

- **Easiest:** run `ONE-KNOB-Series-v1.0.0-Windows-x64-Setup.exe`, tick the plug-ins and formats you want and untick the rest.
- **Just this plug-in:** unzip `AtiNAMatiC-HitZone-v1.0.0-Windows-x64.zip` and copy:

File	Copy to
`AtiNAMatiC-HitZone.vst3`	`C:\Program Files\Common Files\VST3\`
`AtiNAMatiC-HitZone.clap`	`C:\Program Files\Common Files\CLAP\`
`AtiNAMatiC-HitZone.lv2`	`C:\Program Files\Common Files\LV2\`

macOS 10.15 or later (Apple Silicon and Intel)

- **Easiest:** open `ONE-KNOB-Series-v1.0.0-macOS-Installer.pkg`, click **Customize** and tick the plug-ins and formats you want.
- **Just this plug-in:** unzip `AtiNAMatiC-HitZone-v1.0.0-macOS.zip` and copy:

File	Copy to
`AtiNAMatiC-HitZone.vst3`	`/Library/Audio/Plug-Ins/VST3/`
`AtiNAMatiC-HitZone.component` (AU)	`/Library/Audio/Plug-Ins/Components/`
`AtiNAMatiC-HitZone.clap`	`/Library/Audio/Plug-Ins/CLAP/`
`AtiNAMatiC-HitZone.lv2`	`/Library/Audio/Plug-Ins/LV2/`

The plug-ins aren't signed by Apple. If macOS blocks the installer, right-click it and choose **Open**. If you copied files from the zip, clear the quarantine flag in Terminal, e.g. `xattr -cr /Library/Audio/Plug-Ins/VST3/AtiNAMatiC-HitZone.vst3`.

Linux (x86_64 and ARM64)

- **Easiest:** extract `ONE-KNOB-Series-v1.0.0-Linux-x86_64.tar.gz` (or `-Linux-ARM64`) and run `./install.sh`. It asks which plug-ins and formats to install. Use `./install.sh --system` to install for all users.
- **Just this plug-in:** unzip `AtiNAMatiC-HitZone-v1.0.0-Linux-x86_64.zip` (or `-Linux-ARM64`) and copy the `.vst3` folder to `~/.vst3/`, the `.clap` file to `~/.clap/` and the `.lv2` folder to `~/.lv2/`.

Then rescan plug-ins in your DAW.

7. Controls reference

Control	Range	Default	Description
Sweet Spot	0–100 %	50 %	Tonal character + saturation macro (COLD = bright/clean, HOT = warm/driven)
AUTO	On/Off	On	Enables RMS-based automatic input gain
Response	0–100 %	35 %	Auto-gain tracking speed (visible when AUTO is ON)
Gain	–24 to +24 dB	0 dB	Fixed manual input gain (visible when AUTO is OFF)
Mix	0–100 %	100 %	Wet/Dry blend
Output	–24 to +24 dB	0 dB	Final output level trim

All knobs support **double-click to reset to default**.

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8. Internal signal chain

```
 Input
   │
   ├──► [Dry copy for parallel Mix]
   │
   ▼
[Pre-detector HPF @ 35 Hz]  ← prevents subsonic bias in RMS reading
   │
   ▼
[RMS detector]  (60 ms attack / 250 ms release)
[Peak detector] (instant attack / 600 ms release)
[Peak-hold]     (holds, then 1.5 s decay)
   │
   ▼
[Auto-gain / Manual-gain decision]
   → status → UI: Too Cold / Perfect / Too Hot
   │
   ▼
[Input gain (smoothed 30 ms)]
   │
   ▼
[Tilt EQ]
   ├─ Low shelf @ 180 Hz: −2 to +3 dB (Sweet Spot left → right)
   └─ High shelf @ 4.5 kHz: +2 to −2.5 dB (Sweet Spot left → right)
   │
   ▼
[Asymmetric soft saturation]  ← only above 50 % Sweet Spot
   (2nd-harmonic bias, tube-style, blends in progressively)
   │
   ▼
[Output gain (smoothed 20 ms)]
   │
   ▼
[Parallel Mix with dry copy]  (smoothed 20 ms)
   │
   ▼
Output  →  NAM Plugin
```

9. Typical use cases

Standard NAM rig (recommended starting point)

- AUTO: **On** · Sweet Spot: **50 %** · Response: **35 %** · Mix: **100 %** · Output: **0 dB**

Set it and forget it. HitZone centres your signal in NAM's sweet zone and Status will read "Perfect" within a few seconds of playing.

High-gain / tight riff work

- AUTO: **On** · Sweet Spot: **25 %** (COLD side) · Response: **70 %** · Mix: **100 %** · Output: **0 dB**

The COLD tonal character (slight high-shelf lift) keeps chug tight and articulate. Fast Response prevents any level softening between palm mutes.

Clean / crunch with vintage warmth

- AUTO: **On** · Sweet Spot: **70 %** (HOT side) · Response: **30 %** · Mix: **100 %** · Output: **–1 dB**

The HOT tonal character adds low-mid weight and a hint of tube saturation that sits well under cleaner NAM models. Slow Response avoids pumping on sustained notes.

Fixed gain (recording / automation)

- AUTO: **Off** · Gain: set by ear until Status reads "Perfect" · Mix: **100 %** · Output: **0 dB**

Useful when you need a locked gain value — for instance, when recording automation lanes or when your interface's preamp level is already dialled for NAM.

Parallel blend

- AUTO: **On** · Mix: **40–60 %** · Output: **0 dB**

Blends HitZone's conditioned signal with the raw dry signal. Adds level intelligence and tonal shaping while preserving pick attack from the unprocessed chain.

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10. Tips

- **Start with AUTO ON at 50 % Sweet Spot.** HitZone will normalise most guitars to Perfect within 2–3 seconds of playing. Adjust Sweet Spot from there to taste.
- **Watch the Gain Readout.** If AUTO consistently applies more than ±10 dB, your source level is significantly off NAM's ideal. Consider adjusting at the source (DI level, track fader) so HitZone has less work to do.

- **The safety stomp is always on.** Even in Manual mode, the auto system's peak-danger check does not apply — but the status indicator still tells you when you're Too Hot. Trust it.
- **Use COLD for metal, HOT for blues.** The tonal presets are subtle but stack up across a full mix. High-gain drop-tuned riffs sound tighter on the COLD side; single-coil and P-90 players often prefer the HOT side's extra body.
- **Response at 100 % can pump.** Very fast tracking on signals with wide dynamic range (fingerpicking, clean arpeggios) can audibly breathe. Keep Response below 60 % for expressive playing.
- **Zero latency.** HitZone adds no latency — suitable for monitoring through your DAW without buffer compensation.

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11. About

AtiNAMatiC-HitZone — Always Hit the Perfect Level

Designed and developed by **Atij** · Atij 666 Studio

Built with: - JUCE 8 — Raw Material Software Ltd. - VST3 SDK — Steinberg Media Technologies GmbH - Neural Amp Modeler (NAM) — Steven Atkinson & contributors - The open-source audio & guitar-plugin community

Personal and commercial use permitted on machines you own.
Bug reports and feature requests: click the **AtiNAMatiC** title badge in the plug-in window to open the About panel.

AtiNAMatiC-HitZone is part of the AtiNAMatiC Series — a family of focused, hardware-flavoured guitar tools for the modern post-amp signal chain.